higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

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NATIONAL CERTIFICATE

MATHEMATICS N5

(16030175)

2 August 2023 (X-paper)
09:00–12:00

Drawing instruments and scientific calculators may be used.

This question paper consists of 5 pages and a formula sheet of 5 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
MATHEMATICS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show all intermediate steps and simplify where possible.
5. All final answers must be rounded off to THREE decimal places.
6. Questions may be answered in any order, but subsections of questions must be kept together.
7. Use only a black or a blue pen.
8. Sketches must be large, neat and fully labelled.
9. Work neatly.

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QUESTION 1

1.1 Determine the following limits:

1.1.1 $$y = \lim_{x \to 0} \frac{e^{-x^2} - \ln(1+x) - 1}{\cos x - \sin x - 1}$$ (2)

1.1.2 $$y = \lim_{x \to 0} \frac{2^{-x} - 1}{3^{-x} - 1}$$ (2)

1.2 Determine the value(s) of x for which $f(x)$ is discontinuous if

$$y = \frac{x^2 - x + 4}{\ln x}$$ (2)

[6]

QUESTION 2

2.1 Determine the derivative of

$$f(x) = \frac{5}{3x + 2}$$

from first principles. (4)

2.2 Determine $\frac{dy}{dx}$ in each of the following cases:

(Simplification NOT required)

2.2.1 $$y = \sin^5[\tan(\cos x^3)]$$ (5)

2.2.2 $$y = \sqrt{x^3 + \sqrt{x^2 + \sqrt{3 - x}}}$$ (4)

2.3 Calculate $\frac{dy}{dx}$ if $y = \frac{(9-3x)^4}{x^2 \cos 2x}$ with the aid of logarithmic differentiation. (4)

2.4 Determine $\frac{dy}{dx}$ of implicit function, (5)

$$e^{xy} = \log 4x - e^{5y}$$

[22]

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QUESTION 3

3.1 Given: $f(x) = x^3 + x^2 - x + 7$

3.1.1 Determine the coordinates of the turning points of $f(x)$ (3)

3.1.2 Draw a table of x and $f(x)$, where x ranges from $x = -3$ to $x = 3$ (2)

3.1.3 Draw a neat graph of $f(x)$ between these values show the turning points (2)

3.1.4 One root of the equation $x^3 + x^2 - x + 7 = 0$ is close to $-2{,}7$. Use Taylor's/Newton's method ONCE to determine a better approximation of this root (root correct to THREE decimal figures). (3)

3.2 The sum of two positive numbers is 18. One of the numbers is multiplied by the square of the other.

Find the numbers that make this product a maximum. (6)

3.3 Sand is poured onto a surface at $15\,\text{cm}^3/\text{s}$, forming a conical pile of which the base radius is always equal to half of its altitude.

How fast is the altitude of the pile increasing when the pile is 3 cm high? (4)

HINT: $V = \frac{1}{3}\pi r^2 h$

[20]

QUESTION 4

4.1 Determine $\int y\,dx$ in each of the following cases:

4.1.1 $y = \dfrac{\tan x}{\ln(\cos x)}$ (3)

4.1.2 $y = \cos(ax)\cos(bx)$ (3)

4.1.3 $y = \cos^3 x \sin^2 x$ (5)

4.1.4 $y = \dfrac{x^3 + 4x^2 + 6}{x + 1}$ (5)

4.1.5 $y = e^{6x} \sin e^{3x}$

HINT: let u or $f(x) = e^{3x}$ (4)

4.2 Determine $\int y\,dx$ by resolving the integral into partial fractions:

$y = \dfrac{2x + 3}{(x - 2)^2}$ (5)

[25]

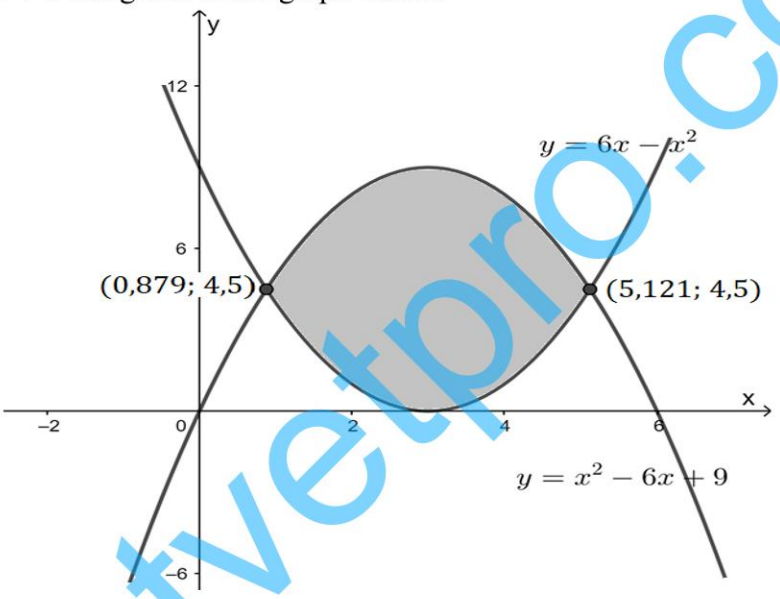
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QUESTION 5

5.1 By using integration, determine the value of the following integral: (6)

$$\int_0^{\infty} xe^{-2x}\,dx$$

5.2 The points of intersection and the area bounded by the curves $y = 6x - x^2$ and $y = x^2 - 6x + 9$ are given in the graph below.



5.2.1 Calculate the magnitude of the area bounded by the two curves. (3)

5.2.2 Calculate the volume of the solid of revolution formed when the area in QUESTION 5.2.1 is rotated about the x-axis. (5)

5.3 Calculate the second moment of area of a rectangular lamina with sides 9 cm and 4 cm about longer side. (5)

[19]

QUESTION 6

6.1 Determine the particular solution of

$$y\frac{dy}{dx} = 3\sqrt{xy^2 + 9x}\,, \qquad \text{if } y(0) = 4$$ (3)

6.2 Determine the particular solution of

$$\frac{d^2y}{dx^2} = 2\left(x - \frac{3}{2}\right)^2 - \frac{1}{2}, \quad \text{if } \frac{dy}{dx}\Big|_{x=0} = 3 \text{ and } y(1) = 2$$ (5)

[8]

TOTAL: 100

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MATHEMATICS N5

FORMULA SHEET

Any other applicable formula may also be used.

TRIGONOMETRY

$$\sin^2 x + \cos^2 x = 1$$
$$1 + \tan^2 x = \sec^2 x$$
$$1 + \cot^2 x = \operatorname{cos} ec^2 x$$

$$\sin 2A = 2\sin A.\cos A$$
$$\cos 2A = \cos^2 A - \sin^2 A$$
$$\tan 2A = \frac{2\tan A}{1 - \tan^2 A}$$

$$\sin^2 A = \frac{1}{2} - \frac{1}{2}\cos 2A$$
$$\cos^2 A = \frac{1}{2} + \frac{1}{2}\cos 2A$$

$$\sin(A \pm B) = \sin A.\cos B \pm \sin B.\cos A$$
$$\cos(A \pm B) = \cos A.\cos B \mp \sin A.\sin B$$
$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 - \tan A.\tan B}$$

$$\sin A.\cos B = \tfrac{1}{2}[\sin(A + B) + \sin(A - B)]$$
$$\cos A.\sin B = \tfrac{1}{2}[\sin(A + B) - \sin(A - B)]$$
$$\cos A.\cos B = \tfrac{1}{2}[\cos(A + B) + \cos(A - B)]$$
$$\sin A.\sin B = \tfrac{1}{2}[\cos(A - B) - \cos(A + B)]$$

$$\tan x = \frac{\sin x}{\cos x}; \sin x = \frac{1}{\operatorname{cos} ecx}; \cos x = \frac{1}{\sec x}$$

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BINOMIAL THEOREM

$$(x+h)^n = x^n + n.x^{n-1}h + \frac{n(n-1)}{2!}x^{n-2}h^2 + \ldots$$

DIFFERENTIATION

$$e = -\frac{f(a)}{f'(a)}$$

$$r = a + e$$

PRODUCT RULE

$$y = u(x).v(x)$$

$$\frac{dy}{dx} = u.\frac{dv}{dx} + v.\frac{du}{dx}$$

$$= u.v' + v.u'$$

QUOTIENT RULE

$$y = \frac{u(x)}{v(x)}$$

$$\frac{dy}{dx} = \frac{v.\frac{du}{dx} - u.\frac{dv}{dx}}{v^2}$$

$$= \frac{v.u' - u.v'}{v^2}$$

CHAIN RULE

$$y = f(u(x))$$

$$\frac{dy}{dx} = \frac{dy}{du}.\frac{du}{dx}$$

$f(x)$	$\frac{d}{dx}f(x)$	$\int f(x)dx$

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ax^n	nax^{n-1}	$\frac{ax^{n+1}}{n+1}+c$
a	0	$ax+c$
e^x	e^x	e^x+c
a^x	$a^x.\ln a$	$\frac{a^x}{\ln a}+c$
$\log_e x$	$\frac{1}{x}$	—
$\log_a x$	$\frac{1}{x\ln a}$	—
$\sin x$	$\cos x$	$-\cos x+c$
$\cos x$	$-\sin x$	$\sin x+c$
$\tan x$	$\sec^2 x$	$\ln(\sec x)+c$
$\cot x$	$-\cos ec^2 x$	$\ln(\sin x)+c$
$\sec x$	$\sec x.\tan .$	$\ln[\sec x+\tan x]+c$
$\cos ecx$	$-\cos ecx.\cot x$	$\ln[\cos ecx-\cot x\}+c$
$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}$	—
$\cos^{-1} x$	$\frac{-1}{\sqrt{1-x^2}}$	—
$\tan^{-1} x$	$\frac{1}{1+x^2}$	—
$\cot^{-1} x$	$\frac{-1}{1+x^2}$	—
$\sec^{-1} x$	$\frac{1}{x\sqrt{x^2-1}}$	—
$\cos ec^{-1}x$	$\frac{-1}{x\sqrt{x^2-1}}$	—

$f(x)$	$\frac{d}{dx}f(x)$	$\int f(x)dx$

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$\frac{1}{\sqrt{a^2-x^2}}$	—	$\sin^{-1}(\frac{x}{a})+c$
$\frac{1}{a^2+x^2}$	—	$\frac{1}{a}\tan^{-1}(\frac{x}{a})+c$
$\frac{1}{x\sqrt{x^2-a^2}}$	—	$\frac{1}{a}\sec^{-1}(\frac{x}{a})+c$
$\sqrt{a^2-x^2}$	—	$\frac{a^2}{2}\sin^{-1}(\frac{x}{a})+\frac{x}{2}\sqrt{a^2-x^2}+c$
$\frac{1}{x^2-a^2}$	—	$\frac{1}{2a}\ln(\frac{x-a}{x+a})+c$
$\frac{1}{a^2-x^2}$	—	$\frac{1}{2a}\ln(\frac{a+x}{a-x})+c$

INTEGRATION

$$\int f(x).g'(x) = f(x).g(x) - \int f'(x).g(x)dx$$

$$\int \frac{f'(x)}{f(x)}dx = \ln f(x) + c$$

$$\int [f(x)]^n.f'(x)dx = \frac{f(x)^{n+1}}{n+1} + c$$

$$\frac{f(x)}{(ax+b)(cx+d)} = \frac{A}{ax+b} + \frac{B}{cx+d}$$

$$\frac{f(x)}{(x+a)^n} = \frac{A}{(x+a)} + \frac{B}{(x+a)^2} + \frac{C}{(x+a)^3} + \ldots + \frac{Z}{(x+a)^n}$$

Applications of integration

AREAS

$$A_x = \int_a^b y dx; \quad A_x = \int_a^b (y_2 - y_1)dx$$

$$A_y = \int_a^b x dy; \quad A_y = \int_a^b (x_2 - x_1)dy$$

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VOLUMES

$$V_x = \pi\int_a^b y^2\,dx; \quad V_x = \pi\int_a^b (y_2{}^2 - y_1{}^2)dx$$

$$V_y = \pi\int_a^b x^2\,dy; \quad V_y = \pi\int_a^b (x_2{}^2 - x_1{}^2)dy$$

SECOND MOMENT OF AREA

$$I_x = \int_a^b r^2\,dA; \quad I_y = \int_a^b r^2\,dA$$

MOMENTS OF INERTIA

Mass = density x volume/M assa = digtheid x volum e

$$M = pV$$

Definition: $I = mr^2$

General: $I = \int_a^b r^2\,dm = p\int_a^b r^2\,dV$

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